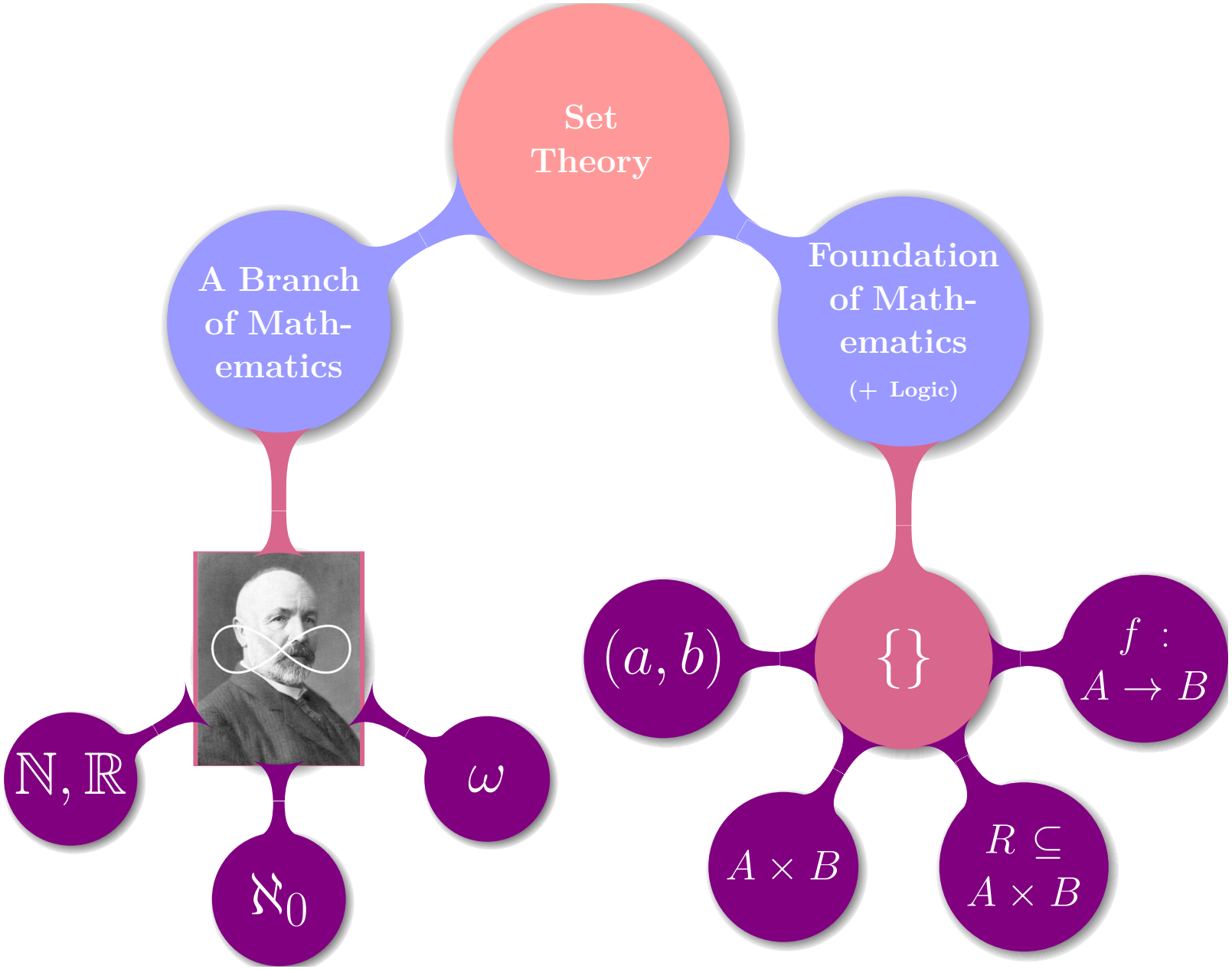
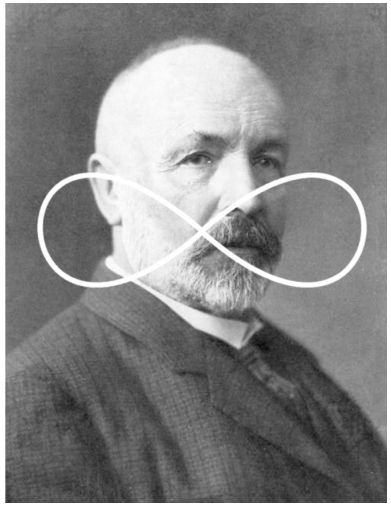
6-infinity: 无穷 (Infinity)

魏恒峰

hfwei@nju.edu.cn

2026 年 04 月 14 日


Set Theory
A Branch of Mathematics
Foundation of Mathematics
(+ Logic)
$\mathbb{N}, \mathbb{R}$
$\aleph_0$
ω
(a, b)
$\{\}$
$f : A \to B$
$A \times B$
$R \subseteq A \times B$

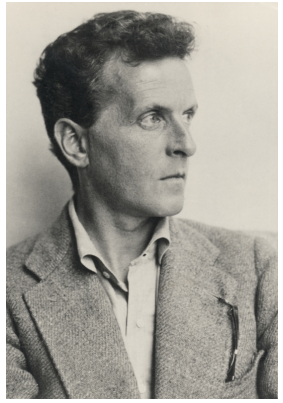


Georg Cantor (1845 – 1918)

David Hilbert (1862 – 1943)

Leopold Kronecker
(1823 – 1891)

Henri Poincaré
(1854 – 1912)



Ludwig Wittgenstein
(1889 – 1951)

From his paradise that Cantor with us unfolded, we hold our breath in awe; knowing, we shall not be expelled.

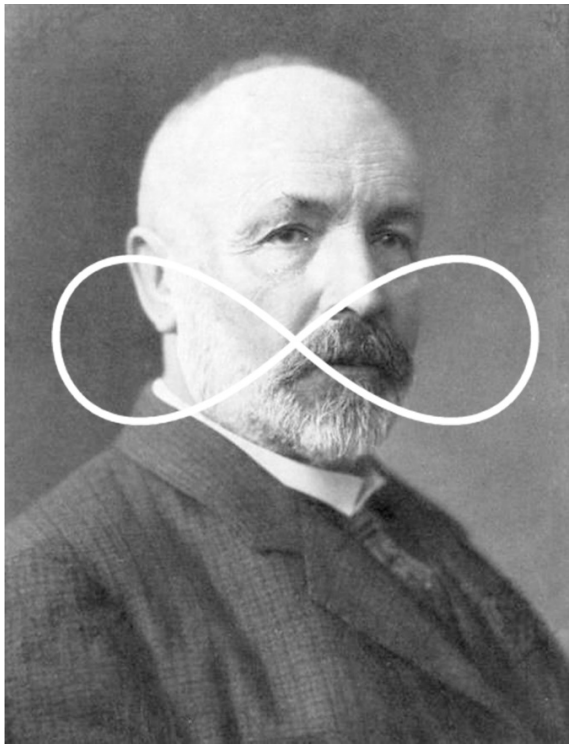
— *David Hilbert*

“没有人能把我们从 *Cantor* 创造的乐园中驱逐出去”

"das wesen der mathematik liegt in ihrer freiheit"

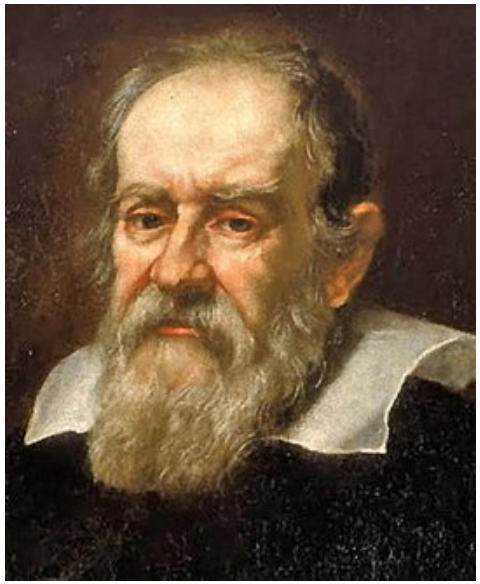
"The essence of mathematics lies in its freedom"

Before Cantor

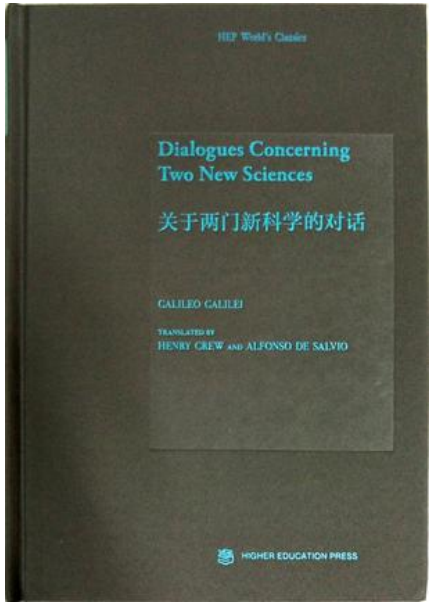


一部上帝安排我们现有空间秩序的方案之书
决定经典
001
一部高度展示人类的逻辑理性与逻辑思维能力的体系教本
全新修订版
[古希腊] 欧几里得 / 著
几何原本
建立空间秩序最久远最权威的逻辑推演语系
Euclid's Elements
13卷图释全本
燕晓东 / 译
凤凰出版传媒集团
江苏人民出版社
凤凰联动 FONGHONG
凤凰决定 DECISION

公理: “整体大于部分”



Galileo Galilei (1564 – 1642)



“关于两门新科学的对话” (1638)

“用我们有限的心智来讨论无限 · · · ”

$$S_1 = \{1, 2, 3, \cdots, n, \cdots\}$$
$$S_2 = \{1, 4, 9, \cdots, n^2, \cdots\}$$

$|S_1| = |S_2|$ $\quad$ $S_2 \subset S_1$

"部分等于全体"

吓得我吃了一鲸

说到底，"等于"、"大于"和"小于"诸性质不能用于无限，而只能用于有限的数量。 — *Galileo Galilei*

无穷数是不可能的。 — *Gottfried Wilhelm Leibniz*

这些证明一开始就期望那些数要具有有穷数的一切性质，或者甚至于把有穷数的性质强加于无穷。

相反，这些无穷数，如果它们能够以任何形式被理解的话，倒是由于它们与有穷数的对应，它们必须具有完全新的数量特征。

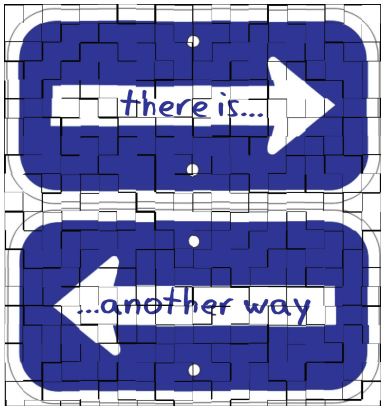
这些性质完全依赖于事物的本性，$\cdots$ 而并非来自我们的主观任意性或我们的偏见。

— *Georg Cantor (1885)*

Definition (Dedekind-infinite & Dedekind-finite (Dedekind, 1888))

A set A is *Dedekind-infinite* if there is a bijective function from A to some proper subset B of A.

A set is *Dedekind-finite* if it is not Dedekind-infinite.



This is a theorem in our theory of infinity.

We have not defined “finite” and “infinite”!

Comparing Sets

$$=$$

$$<$$

Function

[redacted]

[redacted]

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent (等势)* if there exists a *bijection* from A to B.

$$\overline{\overline{A}} \quad \text{(two abstractions)}$$

Abstract from elements: $\{1, 2, 3\} \quad vs. \quad \{a, b, c\}$

Abstract from order: $\{1, 2, 3, \cdots\} \quad vs. \quad \{1, 3, 5, \cdots, 2, 4, 6, \cdots\}$

Definition ($|A| = |B|$ ($A \approx B$) (1878))

A and B are *equipotent* if there exists a *bijection* from A to B.

Q : Is "$\approx$" an equivalence relation?

Theorem (The "Equivalence Concept" of Equipotent)

For any sets A, B, C:

(a) $A \approx B$

(b) $A \approx B \implies B \approx A$

(c) $A \approx B \wedge B \approx C \implies A \approx C$

Do not care too much.

Definition (Finite)

X is finite if

$$\exists n \in \mathbb{N} : |X| = |n| = |\{0, 1, \ldots, n-1\}|.$$

简记为 $|X| = n$

集合 X 是有穷的当且仅当它与某个自然数等势。

Definition (Infinite)

X is infinite if it is not finite:

$$\forall n \in \mathbb{N} : |X| \neq n.$$

Theorem (Existence of Infinite Sets!)

$\mathbb{N}$ *is infinite.* *(So are $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.)*

By Contradiction.

$$\exists n \in \mathbb{N} : |\mathbb{N}| = n.$$

$$\exists f : \mathbb{N} \xleftrightarrow[onto]{1-1} \{0, 1, \cdots, n-1\}$$

$$g \triangleq f|_{\{0,1,\ldots,n\}} : \{0, 1, \cdots, n\} \to \{0, 1, \cdots, n-1\}$$

By the Pigeonhole Principle : g is not 1-1 $\implies$ f is not 1-1

Definition (Infinite)

For any set X,

Countably Infinite

$$|X| = |\mathbb{N}| \triangleq \aleph_0$$

Countable

$$(\text{finite} \vee \text{countably infinite})$$

Uncountable

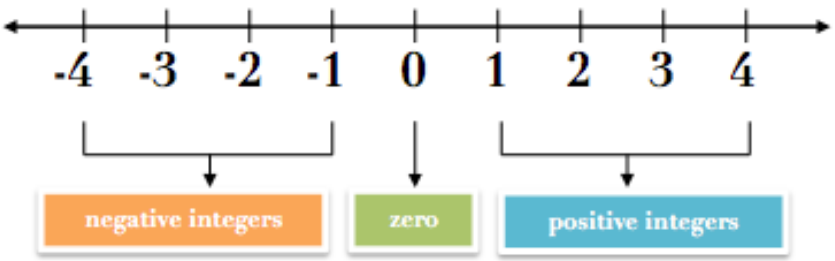
$$(\neg \text{ countable})$$

$$(\text{infinite}) \wedge \Big(\neg\ (\text{countably infinite})\Big)$$

$$\aleph_0$$

Theorem ($\mathbb{Z}$ is Countable.)

$$|\mathbb{Z}| = |\mathbb{N}|$$



$$0 \quad 1 \quad -1 \quad 2 \quad -2 \quad \cdots$$

Theorem ($\mathbb{Q}$ is Countable. (Cantor 1873-11; Published in 1874))

$$|\mathbb{Q}| = |\mathbb{N}|$$

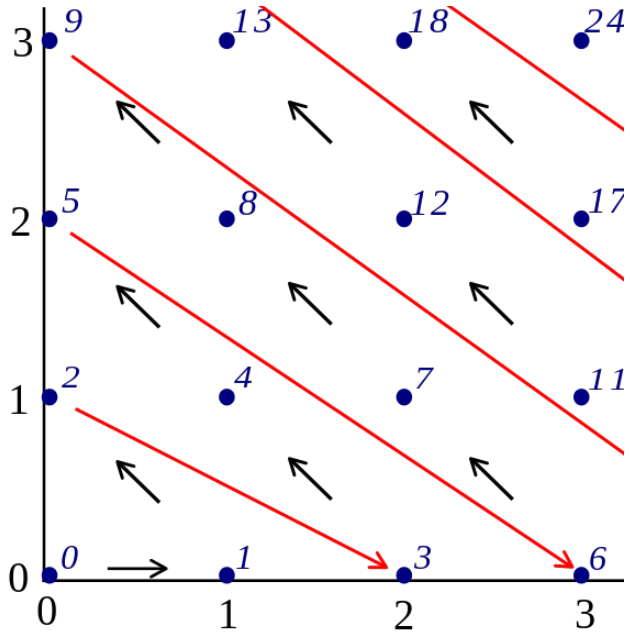
$|\mathbb{Q}| = |\mathbb{N}|$

$$q \in \mathbb{Q}^+ : a/b \ (a, b \in \mathbb{N}^+)$$

1/1	1/2	1/3	1/4	1/5	1/6	1/7	1/8	⋯
2/1	2/2	2/3	2/4	2/5	2/6	2/7	2/8	⋯
3/1	3/2	3/3	3/4	3/5	3/6	3/7	3/8	⋯
4/1	4/2	4/3	4/4	4/5	4/6	4/7	4/8	⋯
5/1	5/2	5/3	5/4	5/5	5/6	5/7	5/8	⋯
6/1	6/2	6/3	6/4	6/5	6/6	6/7	6/8	⋯
7/1	7/2	7/3	7/4	7/5	7/6	7/7	7/8	⋯
8/1	8/2	8/3	8/4	8/5	8/6	8/7	8/8	⋯
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋱

Theorem ($\mathbb{N} \times \mathbb{N}$ is Countable.)

$$|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$$



$$\pi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

$$\pi(k_1, k_2) = \frac{1}{2}(k_1+k_2)(k_1+k_2+1)+k_2$$

$$\pi(2, 1) = 7$$

Cantor Pairing Function

Theorem ($\mathbb{N}^n$ is Countable.)

$$|\mathbb{N}^n| = |\mathbb{N}|$$

Theorem

The Cartesian product of finitely many countable sets is countable.

$$\mathbb{N}^n \quad vs. \quad \mathbb{N}^{\mathbb{N}}$$

$$\pi^{(n)} : \mathbb{N}^n \to \mathbb{N}$$

$$\pi^2(k_1, k_2) = \frac{1}{2}(k_1 + k_2)(k_1 + k_2 + 1) + k_2$$

$$\pi^{(n)}(k_1, \ldots, k_{n-1}, k_n) = \pi(\pi^{(n-1)}(k_1, \ldots, k_{n-1}), k_n) \quad (n \geq 3)$$

Theorem

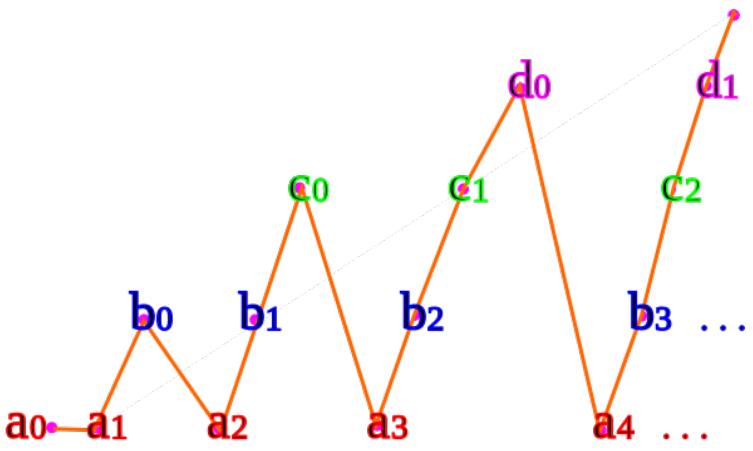
Any finite union of countable sets is countable.

$$A = \{a_n \mid n \in \mathbb{N}\} \quad B = \{b_n \mid n \in \mathbb{N}\} \quad C = \{c_n \mid n \in \mathbb{N}\}$$

$$a_0 \quad b_0 \quad c_0 \quad a_1 \quad b_1 \quad c_1 \cdots$$

Theorem

The union of countably many countable sets is countable.



Counting by Diagonals

Beyond

$$\aleph_0$$

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

Different "Sizes" of Infinity

Cantor's Diagonal Argument (1890)

Theorem ($\mathbb{R}$ is Uncountable. (Cantor 1873-12; Published in 1874))

$$|\mathbb{R}| \neq |\mathbb{N}|$$

By Contradiction.

$$f : \mathbb{R} \xleftrightarrow[onto]{1-1} \mathbb{N}$$

3.14159...
1.41421...
1.73205...
2.23606...
2.71828...
0.14285...

$\Downarrow$

3.43625...

$\Downarrow$

2.32514...

$$\mathfrak{c} \triangleq |\mathbb{R}|$$

$\mathbb{R}$ 是一个连续统 (Continuum)

https://en.wikipedia.org/wiki/Continuum_(set_theory)

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^{n \in \mathbb{N}}|$$

$$f(x) = \tan \frac{(2x-1)\pi}{2}$$

$$|(0,1)| = |(-\frac{\pi}{2}, \frac{\pi}{2})| = |\mathbb{R}|$$

$$\mathbb{R} \times \mathbb{R} \approx \mathbb{R}$$

$$(x = 0.a_1a_2a_3\cdots, y = 0.b_1b_2b_3\cdots) \mapsto 0.a_1b_1a_2b_2a_3b_3\cdots$$

Was Cantor Surprised?

Theorem ($|\mathbb{R}|$ (Cantor 1877))

$$|(0,1)| = |\mathbb{R}| = |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}^n|$$

"Je le vois, mais je ne le crois pas !"

"I see it, but I don't believe it !"

— Cantor's letter to Dedekind (1877).

Q : Then, what is "dimension"?

Theorem (Brouwer (Topological Invariance of Dimension))

There is no continuous bijections between $\mathbb{R}^m$ and $\mathbb{R}^n$ for $m \neq n$.

Beyond

C

Theorem (Cantor's Theorem (1891))

$$|A| \neq |\mathcal{P}(A)|$$

Theorem (Cantor Theorem)

If $f : A \to \mathcal{P}(A)$, then f is not onto.

Theorem (Cantor Theorem)

$$|A| < |\mathcal{P}(A)|$$

$$A \qquad \mathcal{P}(A) \qquad \mathcal{P}(\mathcal{P}(A)) \qquad \ldots$$

There is no largest infinity.

$<$

Definition ($|A| \leq |B|$)

$|A| \leq |B|$ if there exists an *one-to-one* function f from A into B.

Definition ($|A| < |B|$)

$$|A| < |B| \iff |A| \leq |B| \wedge |A| \neq |B|$$

$$|\mathbb{N}| < |\mathbb{R}|$$

$$|X| < |2^X|$$

$$|\mathbb{N}| < |2^{\mathbb{N}}|$$

Definition (Countable Revisited)

X is countable:

$$(\exists n \in \mathbb{N} : |X| = n) \lor |X| = |\mathbb{N}|$$

Theorem (Proof for Countable)

X is countable iff

$$|X| \leq |\mathbb{N}|.$$

X is countable iff there exists a one-to-one function

$$f : X \to \mathbb{N}.$$

Subsets of Countable Set

Every subset B of a countable set A is countable.

$$\exists f : A \xrightarrow{1-1} \mathbb{N}$$

$$g = f|_B$$

Slope

(a) The set of all lines with rational slopes

$$(\mathbb{Q},\ \mathbb{R})$$

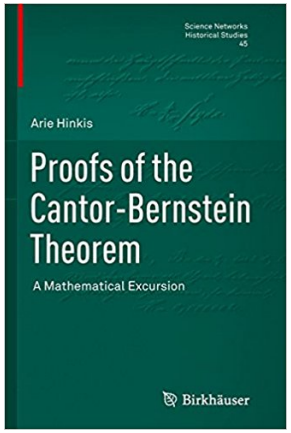
$$|\mathbb{R}| \leq\ |\mathbb{Q} \times \mathbb{R}|\ \leq |\mathbb{R} \times \mathbb{R}| = |\mathbb{R}|$$

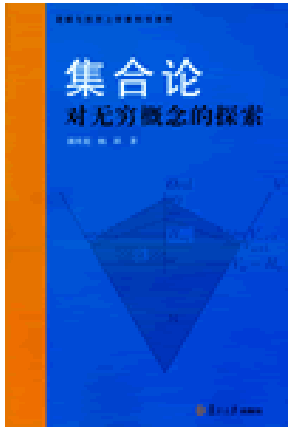
Q : *Is "$\leq$" a partial order?*

Theorem (Cantor-Schröder–Bernstein (1887))

$$|X| \leq |Y| \wedge |Y| \leq |X| \implies |X| = |Y|$$

$$\exists\, f : X \xrightarrow{1-1} Y \wedge g : Y \xrightarrow{1-1} X \implies \exists\, h : X \xleftrightarrow[onto]{1-1} Y$$





Schröder–Bernstein theorem @ wiki

Q : *Is "$\leq$" a total order?*

Theorem (PCC)

Principle of Cardinal Comparability (PCC) $\iff$ *Axiom of Choice*

Theorem

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |\mathcal{P}(\mathbb{Q})|$$

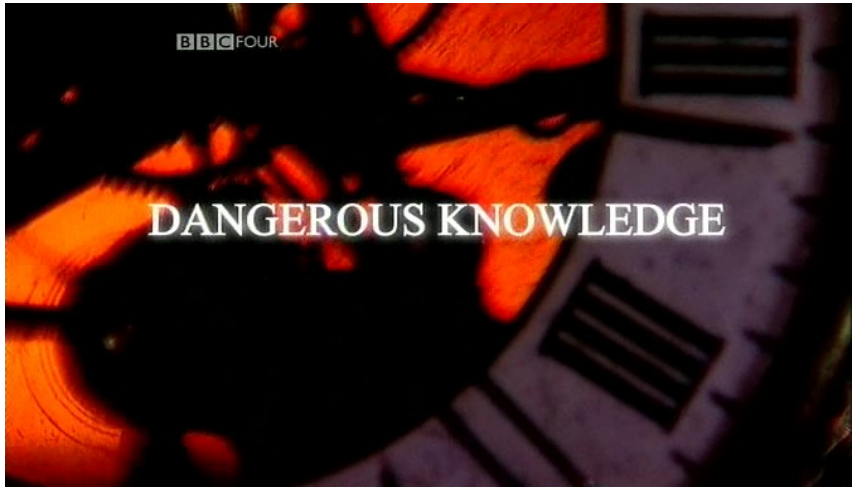
$$|\mathbb{R}| \leq |\mathcal{P}(\mathbb{Q})| \qquad |\mathcal{P}(\mathbb{Q})| \leq |\mathbb{R}|$$

https://en.wikipedia.org/wiki/Cardinality_of_the_continuum#Cardinal_equalities

$$\mathfrak{c} \triangleq |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| = |2^{\mathbb{N}}| \triangleq 2^{\aleph_0}$$

Continuum Hypothesis (CH)

$$\boxed{\nexists A : \aleph_0 < |A| < \mathfrak{c}}$$



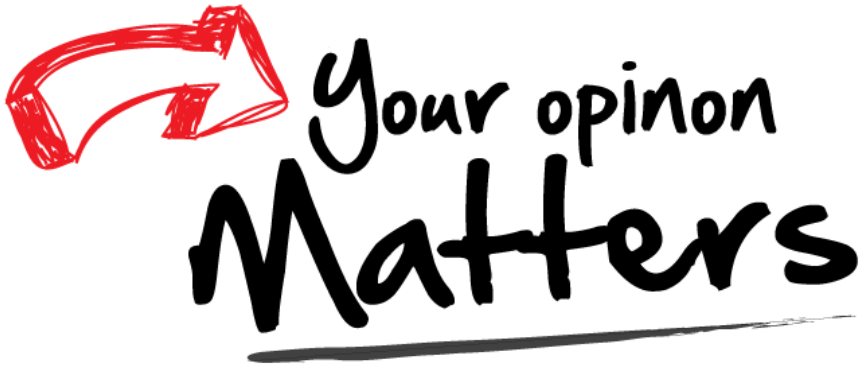
☞ Dangerous Knowledge (22:20; BBC 2007)

Independence from ZFC:

Kurt Gödel (1940) CH cannot be disproved from ZF.

Paul Cohen (1964) CH cannot be proven from the ZFC axioms.

Thank You!



hfwei@hnu.edu.cn